



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING – (24CI3201)

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TITLE OF THE PROJECT	AI-Powered Adaptive Software Testing and Automated Test Coverage Analysis	
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ABSTRACT

AI-Powered Adaptive Software Testing and Automated Test Coverage Analysis is an intelligent software testing platform designed to automate and improve the software testing process using Artificial Intelligence. Traditional testing requires developers and testers to manually create test cases, execute them, identify bugs, and analyze coverage. The proposed system uses AI to automatically analyze source code, identify potential defects, and generate relevant test cases, reducing manual effort and improving testing efficiency.

The system follows an adaptive testing approach, where test cases are prioritized based on code changes, previous test results, and potential risk areas. It automatically identifies important sections of the application that require more testing and focuses testing efforts accordingly. The platform also analyzes test execution results and generates test coverage reports to highlight untested portions of the code.

The proposed system can be integrated with modern CI/CD pipelines using tools such as GitHub Actions and Docker, allowing automated testing whenever new code is committed. By combining AI-based test generation, adaptive test prioritization, automated execution, and coverage analysis, the system aims to improve software quality, detect defects earlier, reduce testing time, and support faster and more reliable software development.

Signature of the Guide

HOD